Type MS · ASCII and Unicode character sets

Answer 1 of 8 · Question 2 part a.iii · 2 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part a.iii in this mark scheme extract

2(a)(i)	Award up to **two** marks for: • 0100 (1) 1110 (1)	Accept any number of leading zeros.	2
2(a)(ii)	**The only correct answer is B** *A is not correct because standard ASCII uses 7 bits, 64 characters would need only 6 bits* *C is not correct because standard ASCII uses 7 bits, 256 characters would need 8 bits* *D is not correct because standard ASCII uses 7 bits, 512 characters would need 9 bits*		1
2(a)(iii)	Award up to **two** marks for a linked explanation such as: All of the major languages/symbols/characters can be represented by Unicode (1) because it uses a minimum of 16 bits/more bits/32 bits/65536 characters (1) Unicode can represent all/more characters/any language (1) whereas ASCII can only represent English/Latin/128 characters/doesn't have enough characters (1) Unicode can represent all characters (1) because it uses 16 bits/2 bytes /more bits instead of 8 bits/1 byte (1)	Accept the reverse argument Allow examples of non-latin characters	2

Question Number	Answer	Additional Guidance	Mark
2(b)	1010 1011 Award up to **two** marks for: • MSB = 1 (1) • Rest of pattern correct 010 1011 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(c)	Award up to **two** marks for: • 0100 (1) 1010 (1)		2

Type MS · ASCII and Unicode character sets

Answer 1 of 8 · Question 2 part a.iii · 2 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part a.iii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(d)(i)</td><td>Award up to two marks for:

• 3 bits used for all patterns (1)/
• No pattern repeated (1)

Example:

<table><tr><th>Colour</th><th>Binary pattern</th></tr><tr><td>Green</td><td>000</td></tr><tr><td>Black</td><td>001</td></tr><tr><td>White</td><td>010</td></tr><tr><td>Red</td><td>011</td></tr><tr><td>Blue</td><td>100</td></tr></table></td><td></td><td>2</td></tr>
<tr><td>2(d)(ii)</td><td>Award one mark for each of:

• 3579 x 6128 x 32 (1)
• ÷ 8 (1)
• + 732 (1)
• ÷ (1000 x 1000) (1)

Examples

$\frac{\frac{3579 \times 6128 \times 32}{8} + 732}{1000 \times 1000}$

$\frac{((3579 \times 6128 \times 32) \div 8) + 732}{1000 \times 1000}$

$\frac{(3579 \times 6128 \times 4) + 732}{1000 \times 1000}$</td><td>• Units are not required
• Equivalent expressions are accepted
• Calculations not explicit but expressed gain the mark
• Award 1 mark for correct calculated answer of 87.73 if no other marks awarded</td><td>4</td></tr>
<tr><td colspan="2"></td><td>Total for question 2</td><td>15</td></tr>
</table>

Type MS · ASCII and Unicode character sets

Answer 1 of 8 · Question 2 part a.iii · 2 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part a.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · ASCII and Unicode character sets

Answer 2 of 8 · Question 2 part c.i · 1 mark

2020 NOV · Paper 01 · extract 1 of 3 · find part c.i in this mark scheme extract

2(a)(i)	1101 1110 Award one mark for: • MSB = 1 (1) • Rest of pattern correct (1)		2

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	- (1) 119 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(b)	B 16		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	128	Allow the range 0-127	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	C		1

Question Number	Answer	Additional Guidance	Mark

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Type MS · ASCII and Unicode character sets

Answer 2 of 8 · Question 2 part c.i · 1 mark

2020 NOV · Paper 01 · extract 2 of 3 · find part c.i in this mark scheme extract

2(c)(iii)	Award **two** marks for a linked explanation such as: (Before Unicode existed) there are hundreds of different encoding systems (1) and no single encoding system could contain enough characters to represent all major languages (1) Standard ASCII only provides 128 different patterns (1) can't represent all major languages/symbols/characters (1) Unicode uses a minimum of 16 bits (1) so can represent at least 2^{16} characters (1) Unicode has very large number of characters (1) can represent all languages/ASCII was developed (just) for English (1)		2

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Type MS · ASCII and Unicode character sets

Answer 2 of 8 · Question 2 part c.i · 1 mark

2020 NOV · Paper 01 · extract 3 of 3 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · ASCII and Unicode character sets

Answer 3 of 8 · Question 2 part c.ii · 1 mark

2020 NOV · Paper 01 · extract 1 of 3 · find part c.ii in this mark scheme extract

2(a)(i)	1101 1110 Award one mark for: • MSB = 1 (1) • Rest of pattern correct (1)		2

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	- (1) 119 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(b)	B 16		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	128	Allow the range 0-127	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	C		1

Question Number	Answer	Additional Guidance	Mark

Type MS · ASCII and Unicode character sets

Answer 3 of 8 · Question 2 part c.ii · 1 mark

2020 NOV · Paper 01 · extract 2 of 3 · find part c.ii in this mark scheme extract

2(c)(iii)	Award **two** marks for a linked explanation such as: (Before Unicode existed) there are hundreds of different encoding systems (1) and no single encoding system could contain enough characters to represent all major languages (1) Standard ASCII only provides 128 different patterns (1) can't represent all major languages/symbols/characters (1) Unicode uses a minimum of 16 bits (1) so can represent at least 2^{16} characters (1) Unicode has very large number of characters (1) can represent all languages/ASCII was developed (just) for English (1)		2

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Type MS · ASCII and Unicode character sets

Answer 3 of 8 · Question 2 part c.ii · 1 mark

2020 NOV · Paper 01 · extract 3 of 3 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · ASCII and Unicode character sets

Answer 4 of 8 · Question 2 part c.iii · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part c.iii in this mark scheme extract

2(a)(i)	1101 1110 Award one mark for: • MSB = 1 (1) • Rest of pattern correct (1)		2

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	- (1) 119 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(b)	B 16		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	128	Allow the range 0-127	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	C		1

Question Number	Answer	Additional Guidance	Mark

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Type MS · ASCII and Unicode character sets

Answer 4 of 8 · Question 2 part c.iii · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part c.iii in this mark scheme extract

2(c)(iii)	Award **two** marks for a linked explanation such as: (Before Unicode existed) there are hundreds of different encoding systems (1) and no single encoding system could contain enough characters to represent all major languages (1) Standard ASCII only provides 128 different patterns (1) can't represent all major languages/symbols/characters (1) Unicode uses a minimum of 16 bits (1) so can represent at least 2^{16} characters (1) Unicode has very large number of characters (1) can represent all languages/ASCII was developed (just) for English (1)		2

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Type MS · ASCII and Unicode character sets

Answer 4 of 8 · Question 2 part c.iii · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part c.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · ASCII and Unicode character sets

Answer 5 of 8 · Question 1 part f · 1 mark

2021 JUN · Paper 01 · extract 1 of 2 · find part f in this mark scheme extract

Number			
1(a)	**The only correct answer is B** ***A*** *is not correct because computers only understand binary* ***C*** *is not correct because Hex values will be held as binary so take up the same space* ***D*** *is not correct because Hex values are held as binary and so take the same time to execute*		1

Question Number	Answer	Additional Guidance	Mark
1(b)	Award **one** mark for each of: • 5 • A	Must be in the correct order	2

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is B** ***A*** *is not correct because this is the number of binary patterns that can be represented by 7 bits* ***C*** *is not correct because this is the number of binary patterns that can be represented by 9 bits* ***D*** *is not correct because this is the number of binary patterns that can be represented by 10 bits*		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each nibble 0110 0101		2

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark from: • 01000 (1) • 1000 (1)		1

Type MS · ASCII and Unicode character sets

Answer 5 of 8 · Question 1 part f · 1 mark

2021 JUN · Paper 01 · extract 2 of 2 · find part f in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(f)	7		1

Question Number	Answer	Additional Guidance	Mark
1(g)(i)	Award **one** mark from: • The smallest block (of colour) in an image (1) • A single dot in a picture (1) • A picture element (1) • A single element of a digital screen (1)		1
1(g)(ii)	Award **one** mark from: • 8 x 9 (1) • 9 x 8 (1)	Accept 72	1
1(g)(iii)	4		1

Question Number	Answer	Additional Guidance	Mark

Type MS · ASCII and Unicode character sets

Answer 6 of 8 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

2(a)(i)	Award up to **two** marks for: • 0100 (1) 1110 (1)	Accept any number of leading zeros.	2
2(a)(ii)	**The only correct answer is B** *A is not correct because standard ASCII uses 7 bits, 64 characters would need only 6 bits* *C is not correct because standard ASCII uses 7 bits, 256 characters would need 8 bits* *D is not correct because standard ASCII uses 7 bits, 512 characters would need 9 bits*		1
2(a)(iii)	Award up to **two** marks for a linked explanation such as: All of the major languages/symbols/characters can be represented by Unicode (1) because it uses a minimum of 16 bits/more bits/32 bits/65536 characters (1) Unicode can represent all/more characters/any language (1) whereas ASCII can only represent English/Latin/128 characters/doesn't have enough characters (1) Unicode can represent all characters (1) because it uses 16 bits/2 bytes /more bits instead of 8 bits/1 byte (1)	Accept the reverse argument Allow examples of non-latin characters	2

Question Number	Answer	Additional Guidance	Mark
2(b)	1010 1011 Award up to **two** marks for: • MSB = 1 (1) • Rest of pattern correct 010 1011 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(c)	Award up to **two** marks for: • 0100 (1) 1010 (1)		2

Type MS · ASCII and Unicode character sets

Answer 6 of 8 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 3 · find part a.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(d)(i)</td><td>Award up to two marks for:

• 3 bits used for all patterns (1)/
• No pattern repeated (1)

Example:

<table><tr><th>Colour</th><th>Binary pattern</th></tr><tr><td>Green</td><td>000</td></tr><tr><td>Black</td><td>001</td></tr><tr><td>White</td><td>010</td></tr><tr><td>Red</td><td>011</td></tr><tr><td>Blue</td><td>100</td></tr></table></td><td></td><td>2</td></tr>
<tr><td>2(d)(ii)</td><td>Award one mark for each of:

• 3579 x 6128 x 32 (1)
• ÷ 8 (1)
• + 732 (1)
• ÷ (1000 x 1000) (1)

Examples

$\frac{\frac{3579 \times 6128 \times 32}{8} + 732}{1000 \times 1000}$

$\frac{((3579 \times 6128 \times 32) \div 8) + 732}{1000 \times 1000}$

$\frac{(3579 \times 6128 \times 4) + 732}{1000 \times 1000}$</td><td>• Units are not required
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Type MS · ASCII and Unicode character sets

Answer 6 of 8 · Question 2 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · ASCII and Unicode character sets

Answer 7 of 8 · Question 3 part c.i · 1 mark

2024 JUN · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

3(a)(i)	Award **one** mark for any of the following: • 256 (1) • 2^8 (1) • 2x2x2x2x2x2x2x2 (1)		1
3(a)(ii)	Award **one** mark for each of the following up to a maximum of **three** marks: • 1920 x 1080 in the numerator (1) • x 16/8 (16 in the numerator and 8 in the denominator) or (2 in the numerator) (1) • / (1024 x 1024) in the denominator (1) 1920 pixels x 1080 pixels x 8 bits / 8 bits per byte / 1024 x 1024 bytes per MiB $\frac{1920 \times 1080 \times \frac{16}{8}}{1024 \times 1024}$	Max 2 marks if not a fully correct expression. Allow equivalence e.g. $1024 = 2^{10}$ $1024^2 = 2^{20}$ 2^{-20} is equivalent to $1/2^{20}$	3
3(a)(iii)	**The only correct answer is C** ***A** is not correct because the MSB is 1 so the denary value must be more than 128* ***B** is not correct because the LSB is 0 so the denary value must be even* ***D** is not correct because the denary number is signed*		1

Type MS · ASCII and Unicode character sets

Answer 7 of 8 · Question 3 part c.i · 1 mark

2024 JUN · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

3(b)(i)	Award **one** mark for each of: • Negative sign (1) • 63 (1) Example: - 63		2
3(b)(ii)	Award **one** mark for: 1000 0010 (1)		1
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • An overflow error has occurred (1) because the result requires more bits to store than are available (1)		2
3(c)(i)	Award **one** mark for any of the following: • 011 0101 (1) • 53 (1) • Hex value of 35 (1) Do not accept 11 0101 (six bits, ASCII is at least seven)		1

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Type MS · ASCII and Unicode character sets

Answer 7 of 8 · Question 3 part c.i · 1 mark

2024 JUN · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract

3(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It is a standard/universal encoding method (1) • It can represent all the characters in the English alphabet (1) • It only uses 7/8 bits **per character** (1) • It can represent 128/256 characters (in standard/extended format) (1) • It takes up less storage space than Unicode (1) • It can be used with bit masking (1) • It is compatible with UTF-8 (1)		2
		Total for question 3	**13**

Type MS · ASCII and Unicode character sets

Answer 7 of 8 · Question 3 part c.i · 1 mark

2024 JUN · Paper 01 · extract 4 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · ASCII and Unicode character sets

Answer 8 of 8 · Question 3 part c.ii · 2 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

3(a)(i)	Award **one** mark for any of the following: • 256 (1) • 2^8 (1) • 2x2x2x2x2x2x2x2 (1)		1
3(a)(ii)	Award **one** mark for each of the following up to a maximum of **three** marks: • 1920 x 1080 in the numerator (1) • x 16/8 (16 in the numerator and 8 in the denominator) or (2 in the numerator) (1) • / (1024 x 1024) in the denominator (1) 1920 pixels x 1080 pixels x 8 bits / 8 bits per byte / 1024 x 1024 bytes per MiB $\frac{1920 \times 1080 \times \frac{16}{8}}{1024 \times 1024}$	Max 2 marks if not a fully correct expression. Allow equivalence e.g. $1024 = 2^{10}$ $1024^2 = 2^{20}$ 2^{-20} is equivalent to $1/2^{20}$	3
3(a)(iii)	**The only correct answer is C** ***A** is not correct because the MSB is 1 so the denary value must be more than 128* ***B** is not correct because the LSB is 0 so the denary value must be even* ***D** is not correct because the denary number is signed*		1

9

Type MS · ASCII and Unicode character sets

Answer 8 of 8 · Question 3 part c.ii · 2 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

3(b)(i)	Award **one** mark for each of: • Negative sign (1) • 63 (1) Example: - 63		2
3(b)(ii)	Award **one** mark for: 1000 0010 (1)		1
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • An overflow error has occurred (1) because the result requires more bits to store than are available (1)		2
3(c)(i)	Award **one** mark for any of the following: • 011 0101 (1) • 53 (1) • Hex value of 35 (1) Do not accept 11 0101 (six bits, ASCII is at least seven)		1

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Type MS · ASCII and Unicode character sets

Answer 8 of 8 · Question 3 part c.ii · 2 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

3(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It is a standard/universal encoding method (1) • It can represent all the characters in the English alphabet (1) • It only uses 7/8 bits **per character** (1) • It can represent 128/256 characters (in standard/extended format) (1) • It takes up less storage space than Unicode (1) • It can be used with bit masking (1) • It is compatible with UTF-8 (1)		2
		Total for question 3	**13**

Type MS · ASCII and Unicode character sets

Answer 8 of 8 · Question 3 part c.ii · 2 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark